



Department of **Mathematics and Natural Sciences**
PHY112 - Principles of Physics II
Assignment 1
Total marks: 15

Question 1

Positive charge of 10.0 pC is spread uni-formly along a thin nonconducting ring of radius 15 cm.
 $1 \text{ pC} = 10^{-12} \text{ C}$.

Find

- (a) the magnitude and (b) direction of the electric field produced at point P, at the height of 3.5 m from the center of the ring. 3
- (c) If we consider the same amount of charge is spread uniformly in a disk of the same radius, what will be the electric field at the height of 2.5 m from the center of the disk and the charge density 3
- (d) If we imagine an infinite sheet of a plane with the same electric field as the electric field found from (c) what will be its charge density? 2

Question 2

A system two charges 5C and -4C resides on the y-axis the first one 7 cm above the origin and the second one is 7 cm below the origin.

- (a) Find the electric field at a distance of $x = 30 \text{ cm}$ from the center (origin) of the charges. 3
- (b) What would be the electric field just at the origin? 2

- (c) If the value of the electric field from option (b) is the same as the electric field for an infinite sheet (infinite plane) what would be the surface charge density